

Lab17 – Security Context

In this exercise, you will create a Pod that defines a security context with different options.

1. Define a Pod named `busybox-security-context` that
 - uses the image `busybox:1.28` for a single container running the command `sh -c sleep 1h`.

```
vagrant@kubernetes-control-plane:~/lab$ vim pod.yaml
vagrant@kubernetes-control-plane:~/lab$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: busybox-security-context
spec:
  containers:
  - name: busybox
    image: busybox:1.28
    command: ["sh", "-c", "sleep 1h"]

vagrant@kubernetes-control-plane:~/lab$
```

2. Add an ephemeral Volume of type `emptyDir`. Mount the Volume to the container at `/data/test`.

```
vagrant@kubernetes-control-plane:~/lab$ vim pod.yaml
vagrant@kubernetes-control-plane:~/lab$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: busybox-security-context
spec:
  containers:
  - name: busybox
    image: busybox:1.28
    command: ["sh", "-c", "sleep 1h"]
    volumeMounts:
    - name: vol
      mountPath: /data/test
  volumes:
  - name: vol
    emptyDir: {}

vagrant@kubernetes-control-plane:~/lab$
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```

- Define a security context that runs the container with user ID 1000, with group ID 3000, and the file system group ID 2000. Ensure that the container should not allow privilege escalation.

```
vagrant@kubernetes-control-plane:~/lab$ vim pod.yaml
vagrant@kubernetes-control-plane:~/lab$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: busybox-security-context
spec:
  securityContext:
    runAsUser: 1000
    runAsGroup: 3000
    fsGroup: 2000
  containers:
  - name: busybox
    image: busybox:1.28
    command: ["sh", "-c", "sleep 1h"]
    volumeMounts:
    - name: vol
      mountPath: /data/test
    securityContext:
      allowPrivilegeEscalation: false
  volumes:
  - name: vol
    emptyDir: {}

vagrant@kubernetes-control-plane:~/lab$
```

- Create the Pod object and ensure that it transitions into the "Running" status.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f pod.yaml
pod/busybox-security-context created
vagrant@kubernetes-control-plane:~/lab$ kubectl get pod
NAME                                READY    STATUS    RESTARTS   AGE
busybox-security-context           1/1      Running   0           33s
vagrant@kubernetes-control-plane:~/lab$
```

- Open a shell to the running container and create a new file named `logs.txt` in the directory `/data/test`. What's the file's user ID and group ID?

```
vagrant@kubernetes-control-plane:~/lab$ kubectl exec -it busybox-security-context -- sh
/ $ cd /data/test
/data/test $ touch logs.txt
/data/test $ ls -l
total 0
-rw-r--r--    1 1000    2000          0 Aug 13 13:28 logs.txt
/data/test $ exit
vagrant@kubernetes-control-plane:~/lab$
```